



Answer Final Exam 2019 2020 Sem 2 Sem 3

CALCULUS - foundation (Universiti Teknologi Malaysia)



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ANSWER FINAL PAST YEAR PAPER

SESSION 20192020 SEMESTER 3

1 (a) i. $A = 6$ or 2

ii. f is not continuous at $x = 4$

(b) $3/2$

2 (a) $-4\pi e^{\cos 4(\pi x)} \cos^3(\pi x) \sin(\pi x)$

(b) $e^{2x} (2x + 3)$

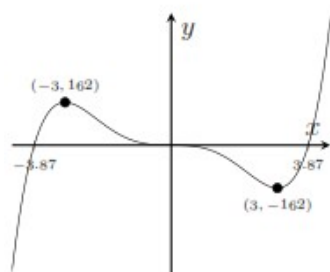
(c) $\frac{dy}{dx} = \frac{y}{\sin x} \left(\frac{3}{2x} - \cos x \ln y \right)$

3 (a) $n = 13$ & Rm 969

(b) (i) ± 3.8730

(b) (ii) inflection point : $(0, 0)$,
minimum point : $(3, -162)$, maximum
point: $(-3, 162)$

(b) (iii)



4 (a) 0.1191

(b) $\frac{x^2}{2} - 3x + 2 \ln x - \ln(x + 1) - \frac{2}{x+1} + C$

5 (a) $a = 1$

(b) 3.6667

(c) 25.2375

6 (a) 27183

(b) $x^{1/2} y = x \sin x + \cos x + C$

7 (a) (i) there is a root (IVT)

(a) (ii) 0.1476

(a) (iii) 0.2652

(b) 12.4028

SESSION 20192020 SEMESTER 2

1 a) 2

b) i. 1 ii. $\sqrt{3}$

2 a) $(1 + x^2)^2 \frac{dy}{dx} + e^{-x}(x + 1)^2 = 0$

b) $-\frac{2}{\sin^2 2t} + \frac{4t \cos}{\sin^3 2t}$

c) $y = 1$ and $dy dx = 0$

3 a) $V = 20$ km per hour

b) i. $y = -x^3 + 5x^2 - 7x + 3$

ii. 0

iii. $(1, 0)$ is minimum & $(7/3, 32/27)$ is
maximum point

iv. sketch

4 a) i. $\frac{1}{2} \left[\frac{\sin 7x}{7} + \sin x \right] + \frac{1}{2} \left[\cos 2x - \frac{\cos^3 2x}{3} \right] + A$

ii. $-\frac{1}{4} e^{-2x} \cos 2x - \frac{1}{4} e^{-2x} \sin 2x + A$

b) $x + e^{-x} - \ln|e^x + 2| + A$

5 a) $4/3$

b) 28.8π

6 a) $y^2 = \ln \left(-\frac{2}{x} + D \right)$

b) $y = \frac{(\ln)^2}{2x} + \frac{3}{x}$

7 a) 1.5748

ii. -1.0000

b) 21.6108

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